B-015

18803

Roll No. _____ Total No. of Pages: 04

18803

BCA I Sem Main Exam Acd. Session 2025-26

BCA-103 – Programming with C Language

Time : 3 Hours

Maximum Marks: 70
Min. Passing Marks:

Instructions to Candidates:

Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All questions are compulsory.

Part- B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. _____ 2. _____

P.T.O.

PART-A

(Short Answer Questions) [10x2=20]

Q.1 What does the return 0; statement in the main function signify?

Q.2 What is the purpose of the #define pre-processor directive in C heacber files?

Q.3 Write the output of the following code:

```
#include <stdio.h>
int main(void)
{
   int a = 5, b = 10;
   printf("%d", printf("Hello "));
   printf("%d", scanf("%d", &a));
   return 0;
}
```

Q.4 Convert the binary number 10110110011 to hexadecimal number.

Q.5 What is bitwise XOR Operator (^) used for in C?

Q.6 How do you pass a 2D array to a function in C?

Q.7 What is the difference between Passing by Value and Passing by reference?

Q.8 What is the purpose of the continue statement inside a loop?

Q.9 What is the difference between *ptr and &ptr in C?

Q.10 What is function in C? Write syntax.

PART-B

(Analytical/Problem Solving Questions) [5x4=20]

Note: Answer any FIVE questions out of SEVEN.

Q.1 How does the fopen() function work in C, and what are the different file modes available? Provide examples for each mode.

Q.2 Write a function that takes an integer pointer, multiplies the value it points to by 2, and returns the updated value.

Q.3 Write a C program to display the following pattern:

```
**********
*        *
*        *
*        *
*        *
*        *
**********
```

Q.4 Create a program that checks if a given character is an uppercase letter, a lowercase letter, or a digit using conditional statements. The program should print the appropriate category for the input character.

Q.5 How does `sizeof()` help when working with dynamic memory allocation in C? Provide an example of its usage.

Q.6 How do relational and logical expressions interact in C? Provide an example where both are used in the same expression to make a complex decision.

Q.7 How can you implement dynamic 2D arrays in C using pointers? Discuss how to allocate memory for a 2D array, access its elements, and deallocate memory correctly.

PART-C

(Descriptive/Analytical/Problem Solving Questions) [3x10=30]

Note: Answer any THREE questions out of FIVE.

Q.1 What is the difference between a pointer and a regular variable in C? How does a pointer store memory addresses and how does this affect variable manipulation?

Q.2 What is an array in C, and how is it declared and initialized? Provide examples of 1D and 2D arrays in C.

Q.3 What are iterative statements in C, and how do they help in automating repetitive tasks? Explain the structure of For, while and do-while loops, and Provide examples for each.

Q.4 What is union in C, and how do they differ from structures? Explain the advantages and limitations of using union to store multiple data types in the same memory location.

Q.5 How does C handle memory management for dynamically allocated variables? Discuss the relationship between malloc(), free() and memory leaks in the context of variable allocation.